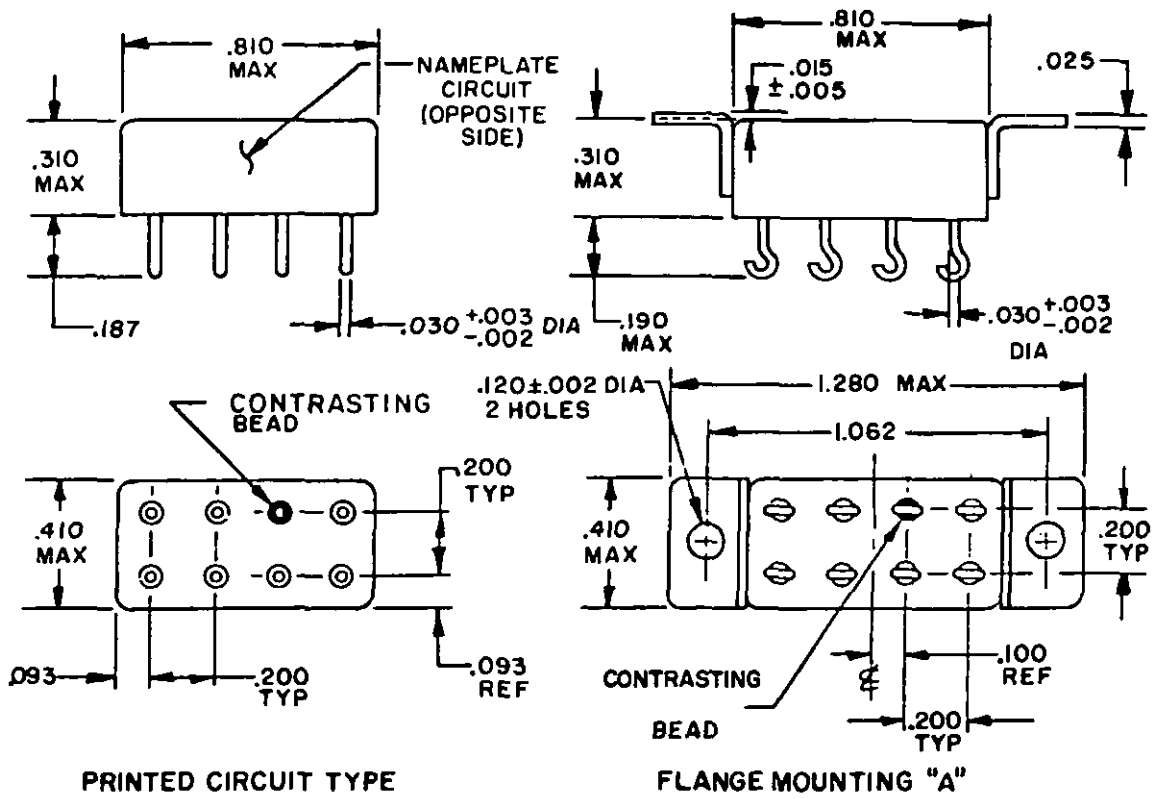
MILITARY SPECIFICATION SHEET

Ⓑ RELAYS, SOLID STATE, REPEAT CYCLE TIMER, TYPE IV, CLASS C, SPST, 250 MILLIAMPERES

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

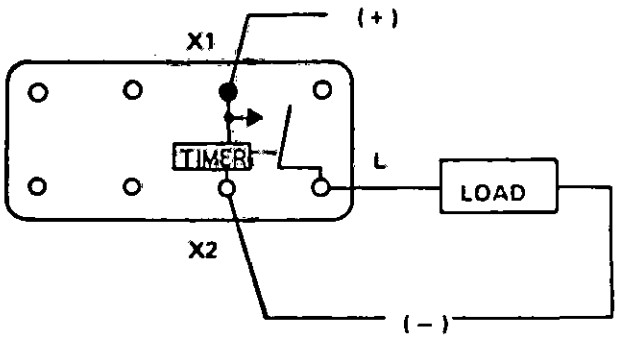
The complete requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-83726.



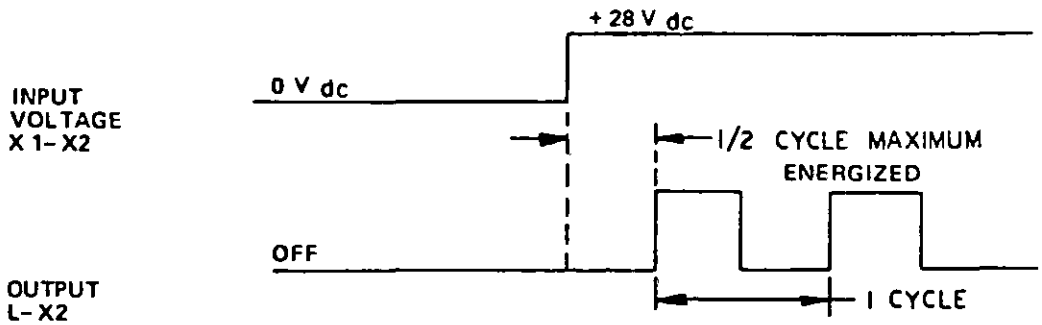
Ⓑ FIGURE 1. Outline dimensions and configuration of relay.

Ⓑ denotes changes

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CIRCUIT DIAGRAM
(SEE NOTE 4)



TIMING DIAGRAM

INCHES	MM	INCHES	MM
.002	0.05	.187	4.75
.003	0.08	.190	4.83
.010	0.25	.200	5.08
.025	0.64	.310	7.87
.030	0.76	.410	10.41
.093	2.36	.810	20.57
.100	2.54	1.062	26.97
.120	3.05	1.280	32.51

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.76 mm) for two place decimals.
4. Spare terminals are connected internally. Do not use for external tie points or for terminals.

Ⓑ FIGURE 1. Outline dimensions and configuration of relay - Continued.

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REQUIREMENTS:

OPERATING DATA:

Timing action: Repeat cycle.

Cycle rate: 10 maximum per second; 1.0 minimum per 60 seconds; on time 50 percent.

Timing accuracy 1/: 10 percent of the total cycle time.

Recycle time 2/: 10 milliseconds.

INPUT DATA:

Input voltage: 28 V dc nominal; range 18 to 32 V dc.

Duty rating: Continuous.

Current drain: 5 milliamperes maximum plus load at 25°C.

Polarity protection: The timer shall be inoperative during, and undamaged by, reversal of the polarity of the operating voltage.

OUTPUT DATA:

Configuration: SPST; switch closure to +28 V dc.

Rating: Equivalent of two MS25237-387 lamps in parallel.

Suppression: Inductive suppression provided for output protection.

Voltage drop: 2 V dc maximum.

Endurance: 1,000,000 cycles minimum for test.
Qualification: 2,000 hours or 1,000,000 cycles whichever is less.
Group B: 1,000 hours or 1,000,000 cycles whichever is less.

ELECTRICAL DATA:

Transients: MIL-STD-704, figure 9, limit 1, duty cycle 2 percent for category B equipment.

Spike:

Self generated: None.

Susceptibility: +80 V maximum; -600 V maximum.

Radio noise: MIL-STD-461, class 1D.

Motor load: Not applicable.

Insulation resistance: 1,000 megohms at 500 V dc at sea level, and 100 V dc at 80,000 feet between each pin and case.

Dielectric strength: 1,000 V rms at 60 Hz at sea level, and 350 V rms at 80,000 feet between case and pins connected together.

1/ The accuracy requirement applies for any combination of operating temperature and voltage.

2/ Recycle time is defined as the maximum time that power must be removed from the input terminals to assure that the next timing cycle will be completed within the specified timing tolerance. (Units can be recycled during timing or after time-out.)

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ENVIRONMENTAL DATA:

Ambient temperature (operating or nonoperating): -55°C to +125°C.

Vibration (sinusoidal): 10 to 80 Hz at 0.06 inch decimal to analog (DA); 80 to 3,000 Hz at 30 G's.

Vibration (random): MIL-STD-202, method 214, condition I, letter J, duration of 30 minutes.

Shock: 1,100 G's for 0.5 millisecond.

Acceleration: 100 G's, any axis.

Seal: MIL-STD-883, method 1014, conditions B and C, except for gross leak, step 2.

Moisture resistance: MIL-STD-202, method 106.

PHYSICAL DATA:

Dimensions and configuration: See figure 1.

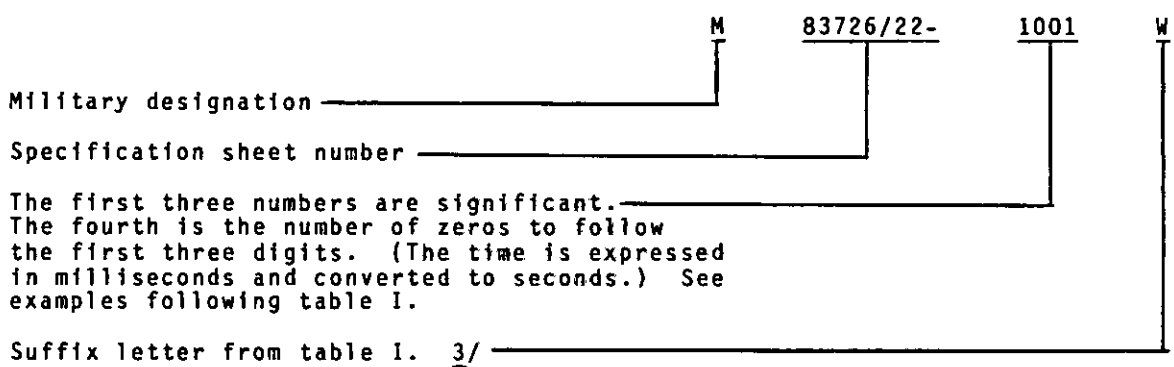
Terminations: See figure 1.

Terminal strength: 3 pounds pull.

Weight: 0.5 ounce.

Ⓑ Marking: Applicable.

Ⓑ PART NUMBER: The part number consists of the prefix MIL-R-83726/22-, a four-digit dash number (denoting the length of one complete cycle (expressed in milliseconds)), and a suffix letter from table I (indicating the extent of EMI requirements and quality level) as follows:



For quality level and mounting style, refer to MIL-R-83726.

Ⓑ 3/ Any relays numbered prior to the date of this specification without a quality level indication shall be considered interchangeable (store and issue) with the W quality level.

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Ⓑ TABLE I. Suffix letter for quality level, printed-circuit mounting and flange mounting.

Quality level	Suffix letters for relays with printed-circuit mounting style	Suffix letters for relays with flange mounting style
M Y X	M Y X	A B C

Examples:

M83726/22-1000M - Time delay repeat cycle 1 per 100 ms with M quality level and printed-circuit mount.

M83726/22-2500A - Time delay repeat cycle 4 per second with M quality level and flange mount.

M83726/22-6002Y - Time delay repeat cycle 1 per 60 seconds with Y quality level and printed-circuit mount.

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 85

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